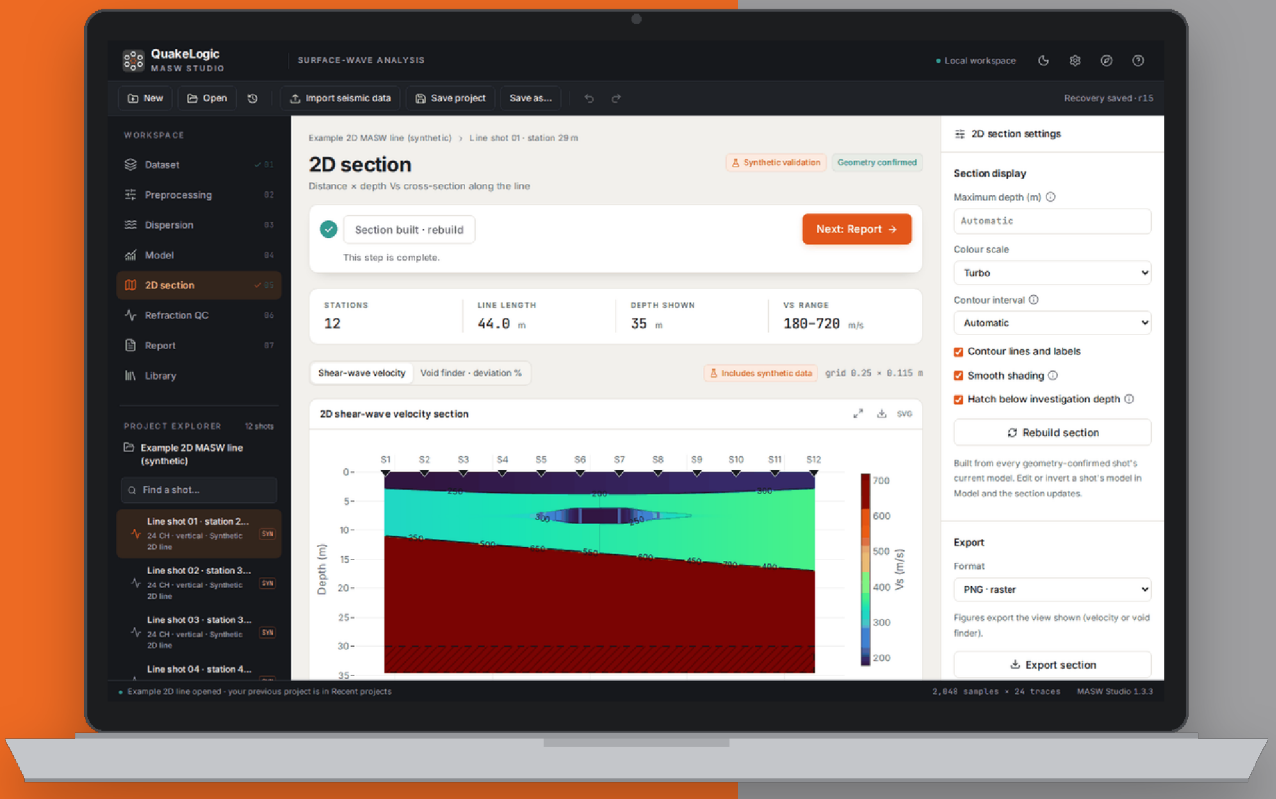
QUAKELOGIC

QUAKELOGIC

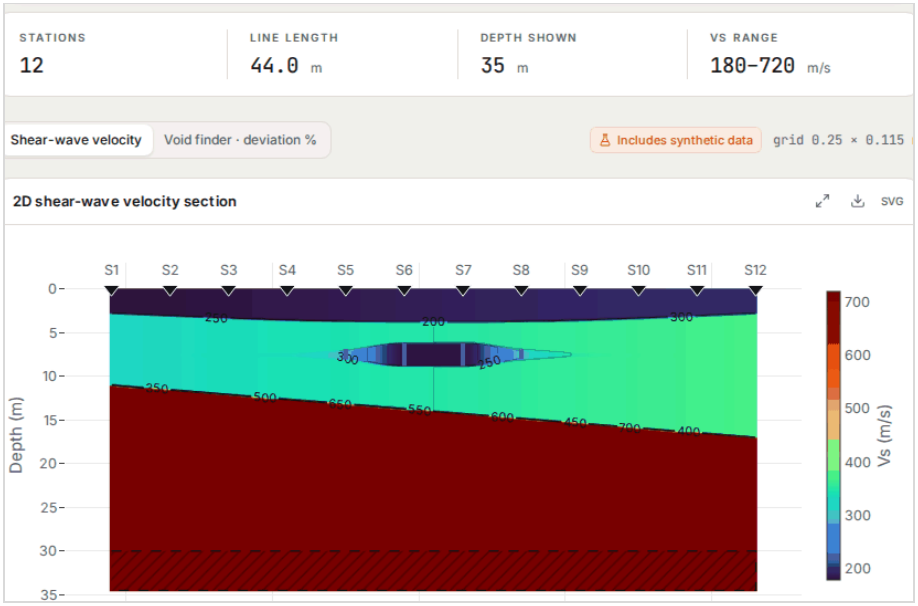
SURFACE-WAVE ANALYSIS SOFTWARE

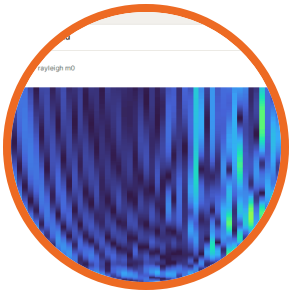
MASW STUDIO



MASW STUDIO

MULTICHANNEL ANALYSIS OF SURFACE WAVES · WINDOWS DESKTOP







FROM FIELD RECORDS TO VS30 AND 2D SECTIONS

QuakeLogic MASW Studio is a standalone Windows application for active-source MASW surveys. It takes multichannel shot records from the seismograph through preprocessing, dispersion imaging and picking, forward modelling and bounded inversion to a layered shear-wave velocity profile, Vs30, a 2D Vs section of the whole line and a technical PDF report. It runs entirely offline, records every processing step, and never modifies the raw field data.

- Reads field data directly: SEG-2, SEG-Y, DRM (.drm) and text exports, recognised from the file contents, a whole survey line at once.
- Defensible results: every step and parameter is recorded in the project, and Vs30 is reported only within the declared interpretation depth.
- The 2D void finder shows each point against its layer average along the line, flagging slower ground, loose fill and possible voids in red.
- Fully offline with no account, activation or telemetry: projects and recovery copies stay on your own computer, ready for remote field sites.

STANDARD FEATURES »

- SEG-2, SEG-Y, DAT, DRM import
- Geometry, units and elevations
- Butterworth filters
- Time and linear mutes
- Phase-shift F–V imaging
- F–K spectrum
- Aliasing and aperture QC
- Click-to-pick named modes
- Rayleigh and Love modelling
- Bounded least-squares inversion
- Vs30 average line (0–30 m)
- 2D Vs section of the line
- 2D section from one long record
- Void finder deviation view
- Refraction first-arrival QC
- PDF report with live preview
- PNG, SVG, PDF and CSV export
- Library of past analyses
- Guided tour and example data

QUAKELOGIC

KEY FEATURES

- **Direct field-data import.** SEG-2 (including 20-bit), SEG-Y, DRM (.drm) and time-column text, with geometry and units read from the headers and confirmed before analysis.
- **Transparent preprocessing.** Detrend, zero-phase Butterworth filters and mutes are previewed beside the original record, then applied as a recorded, removable sequence.
- **Dispersion imaging.** Phase-shift F–V and F–K transforms with phase-coherence normalisation, plus aperture and spatial-aliasing guides showing what the array can resolve.
- **Mode picking.** Click to trace the fundamental and higher modes as named curves, each with explicit uncertainty carried into the inversion.
- **Modelling and inversion.** Rayleigh and Love modes of a layered elastic earth and a bounded, uncertainty-weighted inversion with full convergence history.
- **2D section and void finder.** Station models along a roll-along line become a smooth distance–depth Vs section, with investigation depth hatched and velocity inversions flagged.
- **Reporting.** A branded technical PDF with survey details, site photos, figures and tables, previewed page by page before saving.
- **Library.** Every analysis on the computer is searchable by client, site or notes, and can be reopened or merged into a new project.

PRODUCT FEATURES

Data import
SEG-2 rev 1 · SEG-Y rev 0–2 · DRM (.drm) · CSV/TXT/DAT text · up to 512 channels

Processing
Detrend · Butterworth low/high/band-pass · time and linear mutes · trace edit, stack, split

Dispersion
Phase-shift F–V · F–K · coherence or energy normalisation · acquisition QC guides

Modelling
Rayleigh (P–SV) and Love (SH) modes · multi-mode forward curves · model comparison

Inversion
Bounded trust-region least squares · fixed Vp and density · convergence history

Interpretation
Vs30 travel-time average · declared interpretation depth · 2D Vs and deviation sections · sloping ground

Output
Technical PDF report · PNG/SVG/PDF figures · CSV tables · self-contained .qlmasw projects

For more information about the features, specifications, and available configurations of the MASW STUDIO, please contact QuakeLogic.

SYSTEM REQUIREMENTS & DELIVERY

COMPONENT	CONFIGURATION
Operating system	Windows 10 (1809 or later) or Windows 11, 64-bit
Memory	8 GB RAM recommended
Disk space	About 1 GB
Display	Resizable window; light and dark themes
Internet connection	Not required; runs fully offline
Installer	One installer with bundled runtime
Projects	Self-contained .qlmasw files with raw data and history
Recovery	Automatic local recovery copies, undo and redo
Licence	Perpetual; no activation or licence file
Full version	Every installation; nothing to unlock
Updates	New versions on the QuakeLogic download page
Support	support@quakelogic.net · reply within 3 business days

SUPPORTED FIELD DATA

SPECIFICATION	VALUE
SEG-2	.sg2 · .seg2 · .seg · .dat; 16/32-bit integer, 20-bit, IEEE float
SEG-Y	.sgy · .segy · .dat; IBM/IEEE float, 8/16/32-bit integer
DRM recordings	.drm field recordings with header geometry
Text columns	.csv · .txt · .dat · .asc; time in s or ms
Channels per shot	1 to 512
File size	Up to 256 MiB per record
Samples	Up to 32 million per shot
Geometry units	Metres or feet, converted and confirmed
Survey lines	Multi-shot import for roll-along 2D lines

OPTIONAL FEATURES & ACCESSORIES

PERPETUAL LICENCE

Buy once and use MASW Studio indefinitely. Every installation is the full version: no trial, activation or licence file.

MULTI-SEAT & ACADEMIC

Pricing for several computers, teaching laboratories and research groups, arranged directly with QuakeLogic.

TRAINING & SUPPORT

A narrated training video, the built-in user guide, and email support answered within three business days.

FIELD ACQUISITION PACKAGE

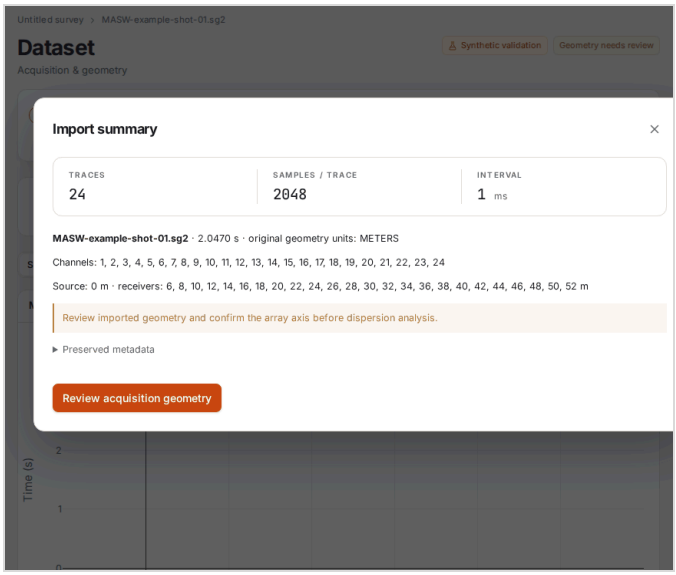
Pair with a QuakeLogic seismograph and geophones for a complete MASW acquisition and analysis package from one supplier.

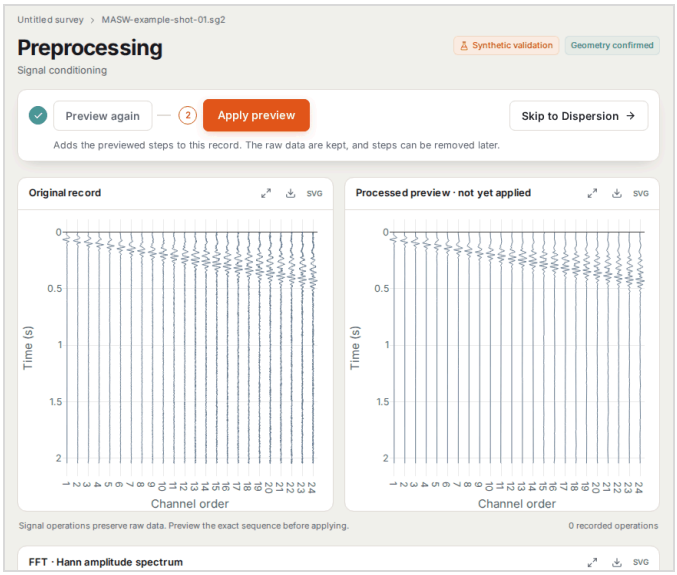
TECHNICAL SPECIFICATIONS

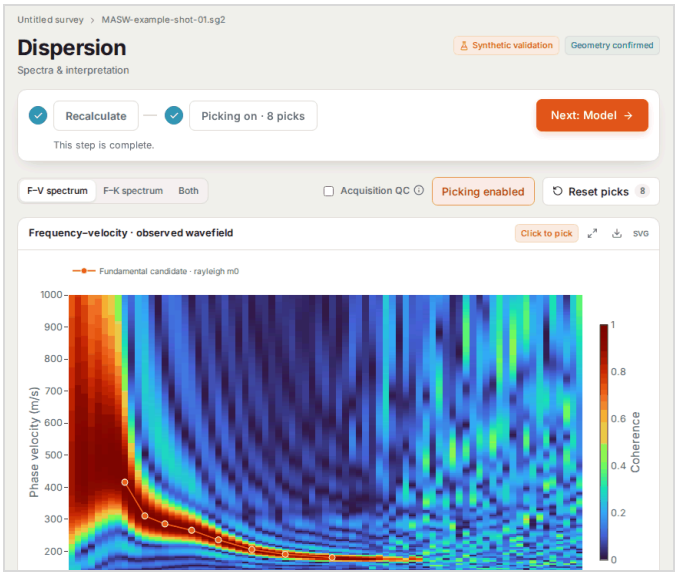
SPECIFICATION	DESCRIPTION
Application	Native Windows desktop window
Architecture	64-bit, bundled runtime
Network use	None; no ports, no cloud
Telemetry	None
Max channels	512 per shot
Max record	256 MiB, 32 million samples
Import formats	SEG-2, SEG-Y, DRM, CSV/TXT/DAT
SEG-2 encodings	Codes 1–5 incl. 20-bit
Filters	Butterworth order 1–10, zero-phase or causal
Mutes	Time window and linear
F–V transform	Phase shift (Park et al. 1999)
Normalisation	Phase coherence or energy
F–K transform	2D spatial Fourier
QC guides	Aperture and spatial aliasing
Picking	Named modes with uncertainty
Wave types	Rayleigh (P–SV), Love (SH)

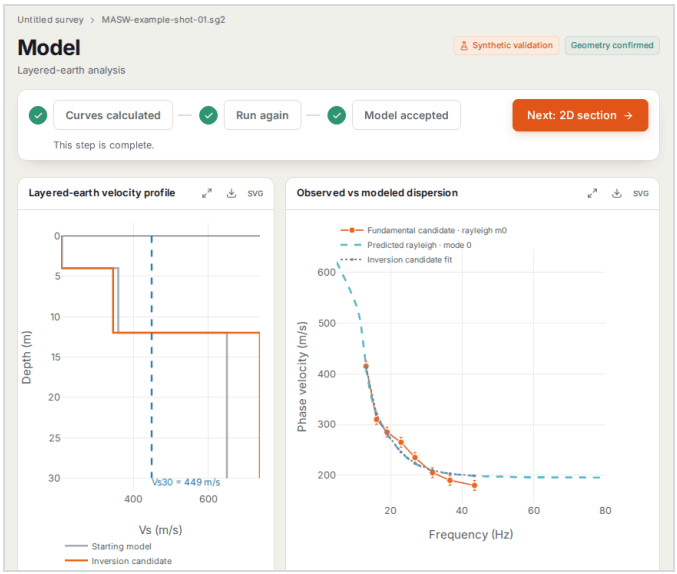
SPECIFICATION	DESCRIPTION
Forward solver	Dunkin / Thomson–Haskell
Inversion	Bounded trust-region least squares
Model	Layered Vs; Vp and density fixed
Vs30	Travel-time average, 0–30 m
2D section	Shape-preserving; elevation view
Void finder	Deviation from layer mean, %
Section QC	Investigation depth hatching
Refraction QC	First-arrival picks and velocities
Report	Branded PDF with page preview
Figure export	PNG, SVG and PDF
Table export	CSV
Project file	.qlmasw, checksummed
History	Undo, redo, recovery copies
Guidance	Guided tour, example SEG-2, user guide
Licensing	Perpetual, no activation
Full version	Every installation

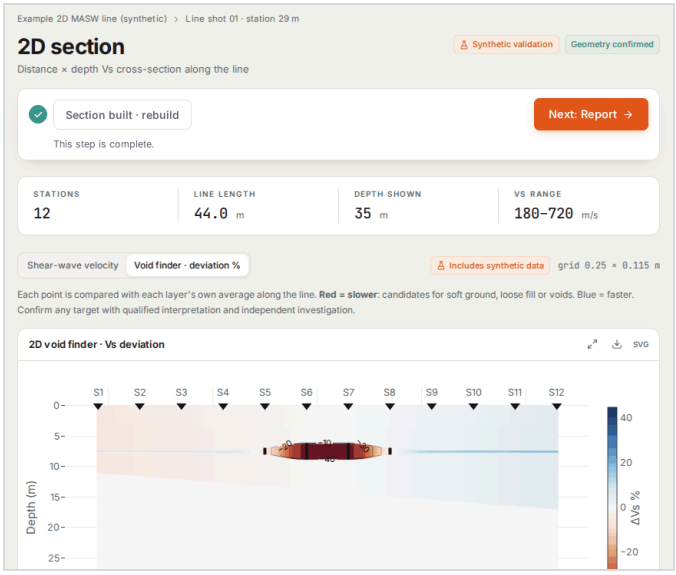
PRODUCT GALLERY

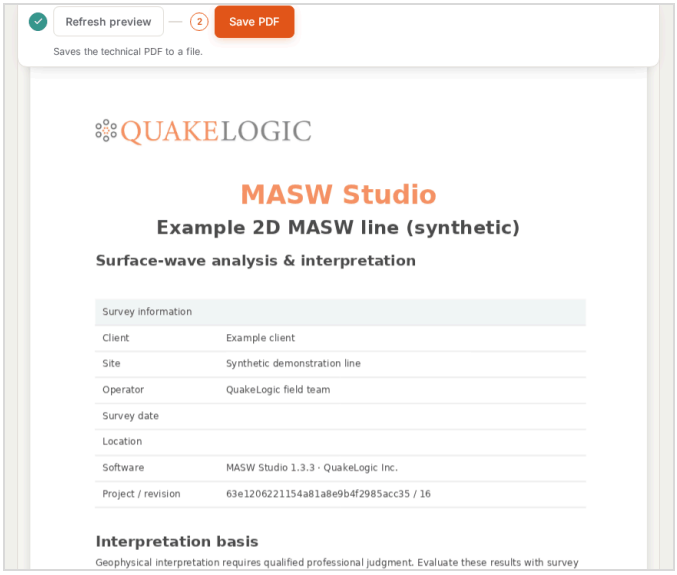






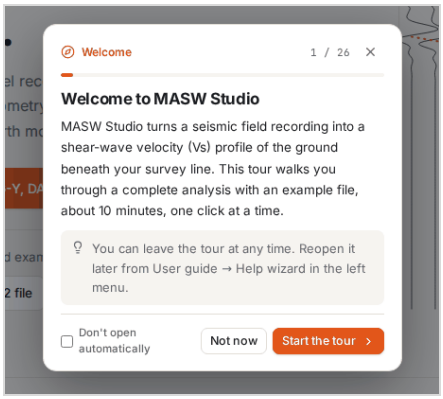




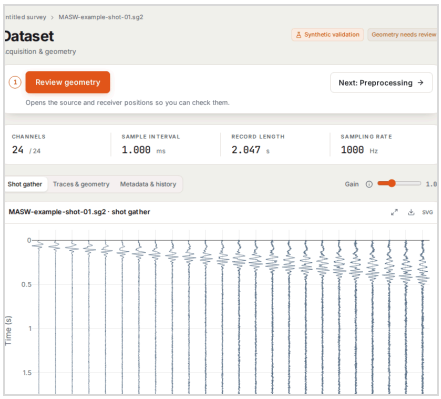


QUAKELOGIC

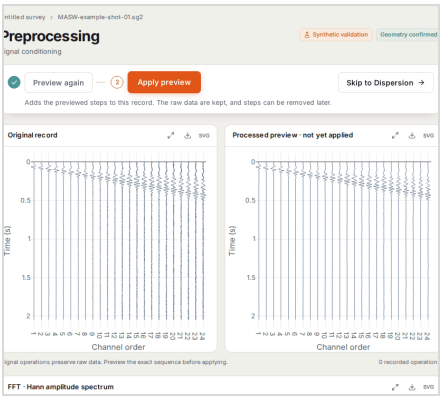
APPLICATION WORKSPACES



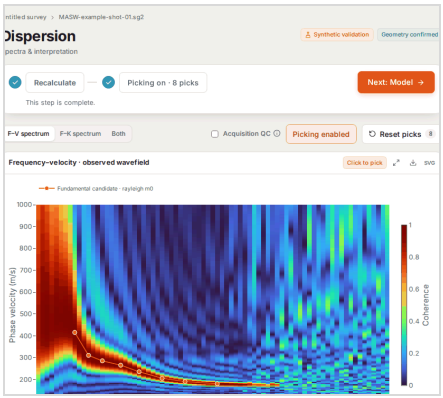
GUIDED TOUR



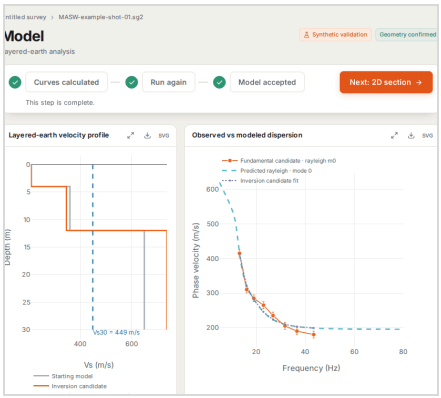
SHOT GATHER



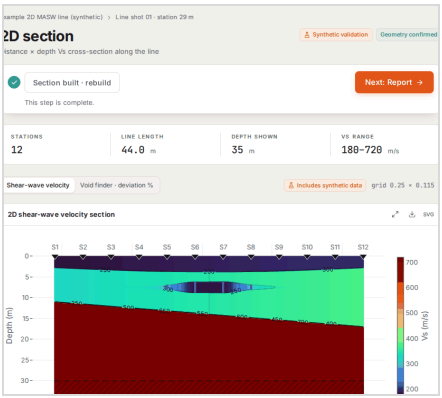
FILTER PREVIEW



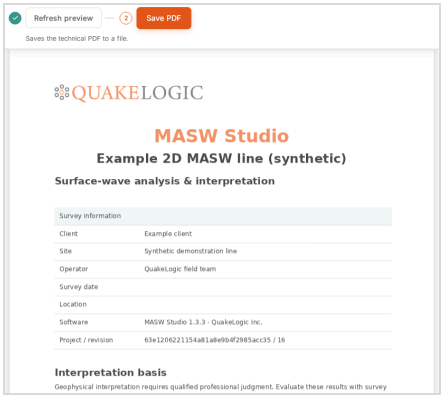
MODE PICKING



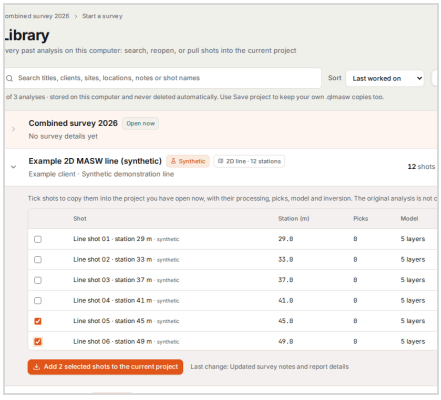
VS PROFILE & VS30



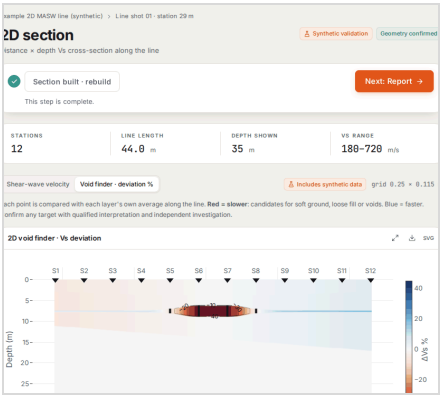
2D VS SECTION



PDF REPORT



ANALYSIS LIBRARY



VOID FINDER

QUAKELOGIC

QUAKELOGIC

MASW STUDIO

Corporate Headquarters:

QUAKELOGIC INC.
2008 Opportunity Dr., Suite 130
Roseville, CA 95678, USA

Monday – Friday
9:00 AM – 5:00 PM (PST/PDT)

Contact Us:

Phone: +1 (916) 899-0391
Email: sales@quakelogic.net
Web: products.quakelogic.net

About QuakeLogic:

QuakeLogic Inc. is a California-based engineering and technology company providing innovative, reliable, high-quality solutions to customers worldwide.

Our Commitment

We are committed to quality, innovation, and customer satisfaction — building lasting partnerships through reliable solutions and dependable support.

Scan to View Product:

